BUSINESS OBJECTIVE

CAPSTONE PRESENTATION

After the COVID-19 pandemic, shifts in consumer behavior—especially the demand for fast, secure, and efficient digital services—have increased the risk of customer churn in the banking industry. ***This project aims to analyze the key factors influencing customer churn*** and develop a predictive model to help banks improve service quality, retain customers, and reduce future acquisition costs.

The bank aims to reduce customer churn in order to retain its existing customer base and enhance its competitive advantage.

Analyze key factors that influence customer churn, such as complaints, age, gender, and other relevant variables.

Use the insights obtained to improve services in alignment with customer needs and to design proactive strategic plans for customer retention.

DATA MINING OBJECTIVE

Develop a model to detect and predict which customers are likely to churn using the following data mining techniques :

1

Build a predictive model to identify which customers are likely to churn.

In this project, Primarily used classification techniques, such as Decision Tree, Random Forest, and Neural Network.

2

Analyze the key factors influencing customer decisions, such as Age, Satisfaction scores, and Complains.

Based on the feature importance results, It was found that Complain accounts for 82.89% of the importance.

3

Compare different models such as Decision Tree, Random Forest, etc. to identify the most accurate one.

The initial evaluation used Accuracy, Precision, Recall, and the Confusion Matrix to assess how well the model

DATASET DESCRIPTION

The target variable is *"Exited"*, which indicates whether a customer is still using the bank's services (0) or has already discontinued them (1).

"Customer Churn Records"

10,000 columns , 18 rows.

- *Personal Information:* Gender, Age, Country
- Service Usage Behavior: Complaints (Complain), Active Membership Status (IsActiveMember), Satisfaction Score
- Financial Information: Account Balance, Number of Products Used, Estimated Income

During the Data Understanding phase, an initial exploration revealed the following insights:

- "*Female*" customers tend to have a higher likelihood of exiting compared to male customers.
- There is a clear correlation between customers who "*Complain*" and those who "*Exited*" the system.